Darshan UNIVERSITY

योग: कर्मसु कौशलम्

# Python Programming - 2301CS404

## Lab - 15 (1)

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### 01) Write a Program to create a class by name Students, and initialize attributes like name, age, and grade while creating an object.

In [1]:
```
class Students:
    def __init__(self,name,age,grade):
        self.name = name
        self.grade = grade
        self.age = age
```

In [3]:
```
s1 = Students('Aditya',19,'A+')
print(s1.name)
```

```
Aditya
```

## 02) Create a class named Bank_Account with Account_No, User_Name, Email,Account_Type and Account_Balance data members. Also create a method GetAccountDetails() and DisplayAccountDetails(). Create main method to demonstrate the Bank_Account class.

In [8]:
```
class Bank_Account:
    def __init__(self,Account_No,User_Name,Email,Account_Type,Account_Balance):
        self.Account_No = Account_No
        self.User_Name = User_Name
        self.Email = Email
        self.Account_Type = Account_Type
        self.Account_Balance = Account_Balance
    def DisplayAccountDetails(self):
        print(self.Account_No,self.User_Name,self.Email,self.Account_Type,self.Account_Balance)
```

In [9]:
```
def GetAccountDetails():
    Account_No = input("Enter AccountNo")
    User_Name = input("Enter User_Name")
    Email = input("Enter Email")
    Account_Type = input("Enter Account_Type")
    Account_Balance = input("Enter Account_Balance")
    return Account_No,User_Name,Email,Account_Type,Account_Balance
details = GetAccountDetails()
b1 = Bank_Account(details[0],details[1],details[2],details[3],details[4])
b1.DisplayAccountDetails()
```

```
123 Aditya asdfghj asdfghjk 55252
```

## 03) WAP to create Circle class with area and perimeter function to find area and perimeter of circle.

In [11]:
```
import math
class Circle:
    def __init__(self,r):
        self.radius = r
    def area(self):
        return math.pi*(self.radius**2)
    def perimeter(self):
        return 2*math.pi*self.radius
```

```
c1 = Circle(10)
print(c1.area())
print(c1.perimeter())
```

```
314.1592653589793
62.83185307179586
```

## 04) Create a class for employees that includes attributes such as name, age, salary, and methods to update and display employee information.

In [14]:
```
class Employee:
    def __init__(self,name,age,salary):
        self.name = name
        self.age = age
        self.salary = salary
    def update(self):
        self.name = input("Enter Updated Name")
        self.age = int(input("Enter Updated Age"))
        self.salary = int(input("Enter Updated Salary"))
    def display(self):
            print(self.name,self.age,self.salary)
e1 = Employee('Aditya',19,25000)
e1.update()
e1.display()
```

```
Aditya 20 250000
```

## 05) Create a bank account class with methods to deposit, withdraw, and check balance.

In [17]:
```
class bank:
    def __init__(self,money):
        self.money = money
    def deposit(self):
        amt = int(input("Enter Amount"))
        self.money+=amt
    def withdraw(self):
        amt = int(input("Enter Amount"))
        if (self.money-amt) < 0 :
            print('Wrong Input')
        else:
            print("Successful")
```

```
            self.money-=amt
    def check_balance(self):
        print(self.money)
b1 = bank(100)
b1.deposit()
b1.withdraw()
b1.check_balance()
```

```
Successful
90
```

## 06) Create a class for managing inventory that includes attributes such as item name, price, quantity, and methods to add, remove, and update items.

In [30]:

```
class inventory:
    ls=[]
    def __init__(self,name,price,qty):
        inventory.ls.append(item(name,price,qty))
        # self.name=name
        # self.price=price
        # self.qty=qty
    def add(self,name,price,qty):
        i1=item(name,price,qty)
        inventory.ls.append(i1)
    def remove(self):
        idx=int(input("enter index"))
        inventory.ls.remove(inventory.ls[idx])
    def update(self):
        search=input("name of inventory")
        for i in inventory.ls:
            if i.name==search:
                i.name=input("enter name")
                i.price=int(input("enter price"))
                i.qty=int(input("enter quantity"))
                break
class item:
    def __init__(self,name,price,qty):
        self.name=name
        self.price=price
        self.qty=qty
```

```
i1=inventory('laptop',50000,5)
i1.add('mobile',30,3)
i1.remove()
print(inventory.ls[0].name)
```

```
mobile
```

## 07) Create a Class with instance attributes of your choice.

In [29]:
```
class Students:
    def __init__(self,name,age,grade):
        self.name = name
        self.grade = grade
        self.age = age
s1 = Students('Aditya',19,'A+')
print(s1.name,s1.age,s1.grade)
```

```
Aditya 19 A+
```

## 08) Create one class student_kit

### Within the student_kit class create one class attribute principal name ( Mr ABC )

### Create one attendance method and take input as number of days.

### While creating student take input their name .

### Create one certificate for each student by taking input of number of days present in class.

In [28]:
```
class student_kit:
    principal = 'Mr ABC'
    def __init__(self,name):
        self.name = name
    def attendence(self):
        self.nod = int(input("Enter No.days"))
    def certificate(self):
        print(self.name,self.nod)
s1 = student_kit('aditya')
```

```
s1.attendence()
s1.certificate()
```

```
aditya 50
```

## 09) Define Time class with hour and minute as data member. Also define addition method to add two time objects.

In [25]:
```
class Time:
    def __init__(self,h,m):
        self.h = h
        self.m = m
    def add(self):
        mint = int(input("Enter Minutes"))
        self.m+=mint
        while(self.m>=60):
            self.h+=1
            self.m-=60
        print(self.h,self.m)
```

In [26]:
```
t1 = Time(8,50)
```

In [27]:
```
t1.add()
```

```
9 10
```

In [ ]: